Myungsin Cho

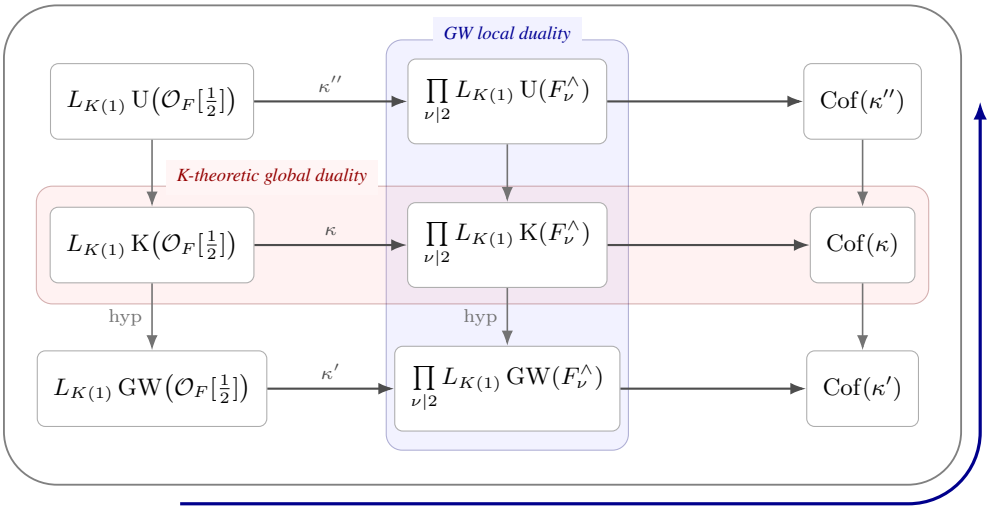
Research Statement

My research is in stable homotopy theory, with connections to algebra and number theory. One direction studies duality in algebraic K-theory, with the long-term goal of understanding duality and descent in Real algebraic K-theory. A second direction concerns equivariant stable homotopy theory and equivariant algebra, where incorporating symmetry leads to new algebraic structures. These interests also provide starting points for undergraduate research.

Duality in algebraic K-theory

In linear algebra, a vector space and its dual, the space of linear functions on it, are isomorphic under suitable finiteness conditions. My research seeks similar phenomena in algebraic K-theory. This theory studies linear algebra over rings R and packages information about general linear groups $GL(R)$ into topological invariants. Although these invariants are difficult to compute, duality can reveal how apparently different pieces of information determine one another.

My dissertation studies this phenomenon by comparing arithmetic information for a number field F as a whole with information obtained at individual primes. In K-theory, this comparison involves three objects: the global theory $L_{K(1)}\,\mathrm{K}\big(\mathcal{O}_F[\tfrac{1}{2}]\big)$, the local theory $\prod_{\nu|2} L_{K(1)}\,\mathrm{K}(F_\nu^\wedge)$, and the comparison term $\mathrm{Cof}(\kappa)$, which measures their difference. Duality reverses this comparison. The global theory and the comparison term exchange roles, while the local theory is self-dual. I establish this symmetry at the prime 2 [Cho25], completing the K-theoretic Tate–Poitou duality previously proved at odd primes by Blumberg and Mandell [BM20]. The red row of the diagram below displays this relationship.



Anderson duality: 180° rotation

A bilinear form, such as an inner product, equips a vector space with additional structure. Grothendieck–Witt theory extends ordinary K-theory by keeping track of these forms. With Lucy Yang, I establish a local duality relating K-theory, GW-theory, and the comparison theory U associated to the hyperbolic map $\mathrm{K} \to \mathrm{GW}$ [CYon]. This duality gives the symmetry shown in the blue column above. Our ongoing work on global fields asks whether this local symmetry and the duality from my dissertation combine into a half-turn symmetry of the full diagram.

Real algebraic K-theory incorporates symmetry to bring ordinary K-theory and GW-theory into a single theory [HM13]. My long-term goal is to understand the dualities above as manifestations of a single duality in this theory and to explain why it exists. In ordinary K-theory, descent connects the theory to arithmetic cohomology, where classical duality provides such an explanation. A corresponding explanation incorporating symmetry is not yet available. I use the known and expected dualities to guide the construction of a Real descent theory that could explain them.

Trace methods connect this duality program to other invariants in topology. The cyclotomic trace relates algebraic K-theory to topological cyclic homology, TC. For the sphere spectrum $\mathbb{S}$, a basic building block of stable homotopy theory, my theorem describes the information missed by this map as a dual of the arithmetic K-theory of the integers [Cho25]. Clausen's work places TC itself within local arithmetic duality, relating it directly to local K-theory and its dual [Cla17].

These connections motivate my study of Real trace methods, which incorporate symmetry into these relationships. With Gerhardt, Keenan, Moreno, and Quigley, I develop homological methods that account for both symmetry and its additional algebraic operations [CGK+26]. We obtain explicit results for Real bordism theories, including $MU_{\mathbb{R}}$ and $BP_{\mathbb{R}}$, extending the methods of Bruner and Rognes to the Real setting. Our next goal is to establish Real Segal conjectures for these theories. I aim to use these methods to investigate how Real topological cyclic homology, TCR, fits into duality in Real algebraic K-theory beyond arithmetic examples.

Equivariant stable homotopy theory and algebra

Symmetry also changes familiar constructions in algebra. For example, the sum and product of a complex number and its conjugate are both real, so they are fixed by complex conjugation. These constructions illustrate how addition and multiplication can combine elements related by symmetry to produce invariant quantities. In equivariant algebra, they develop into operations called transfers and norms. For larger groups, there are operations associated with different subgroups. My research asks which transfers and norms can be used together and how they must interact.

Transfer systems provide a combinatorial tool for studying these operations, [Rub21]. By recording allowed operations as relations between subgroups, they turn questions about equivariant algebra into combinatorial problems. With Chan, Mehrle, Sanchez Ocal, Osorno, Szczesny, and Verdugo, I use this description to study the compatibility of transfers and norms [CCM+25]. As with ordinary addition and multiplication, these operations must obey rules resembling a distributive law. We ask whether choices satisfying these rules can be realized by operads that organize coherent systems of operations.

pairing of N_∞-operads $\Longrightarrow$ compatible pair of transfer systems

$\Longleftarrow$ (partially)

Our results realize several classes of compatible choices, and we conjecture that compatibility is sufficient in general. This would give a combinatorial criterion for when prescribed transfers and norms can form a coherent system of operations.

With Noah Wisdom, I study the algebraic structures carried by invariants in equivariant topology. We ask what conditions allow algebraic data, equipped with representation gradings and equivariant operations, to be realized topologically. For the refined C_2-gradings we consider, we show that the additional grading symmetries are already determined by the existing algebraic operations. This reduces the data that must be specified independently when investigating topological realization.

Undergraduate research I view undergraduate research as a process in which students formulate questions from familiar mathematics and pursue the ideas needed to answer them. Students with a background in linear algebra can already ask meaningful questions by varying a definition, examining new examples, or combining familiar structures. Investigating their own questions gives them a reason to learn more advanced concepts and methods, allowing research to introduce them naturally to ideas from upper-level courses.

My research offers several starting points for this process. Successful undergraduate collaborations on transfer systems illustrate how accessible combinatorial questions can lead to contributions in equivariant topology. Equivariant rings provide opportunities to combine familiar algebra with group action and investigate how ordinary ring-theoretic properties change. The structures and invariants of manifolds with symmetry offer related possibilities for students with geometric interests. These examples would support students in developing questions of their own. I would help them refine their observations into tractable problems and introduce further mathematical tools as their investigations call for them.

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